

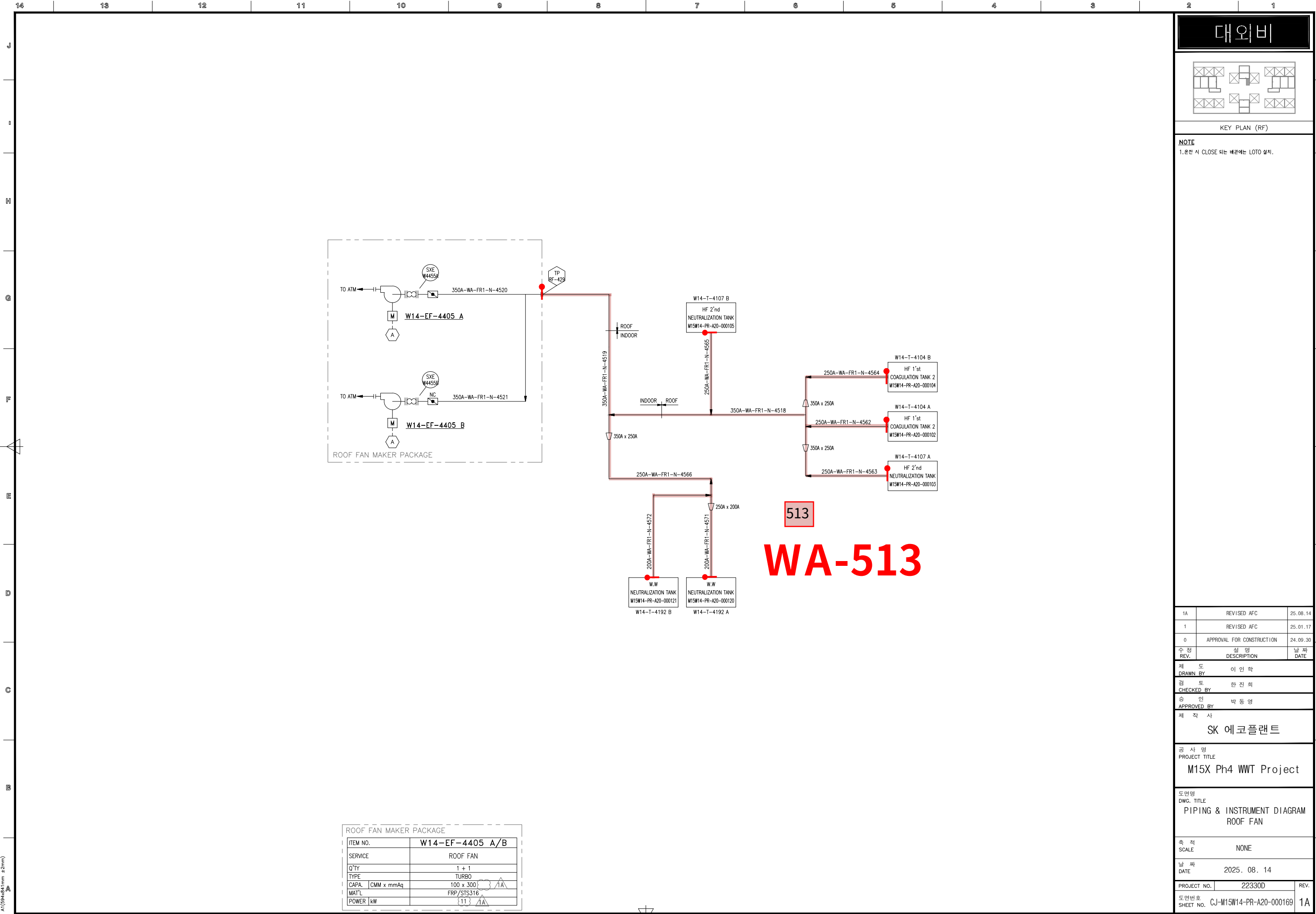
DRAWING LIST FOR TEST PACKAGE

WA-513

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	513	3FA	250A-WA-FR1-N-4566-01	1 OF 1	0	CJ-M15W14-PR-A20-169		
2	513	4FA	250A-WA-FR1-N-4566-01	1 OF 1	0	CJ-M15W14-PR-A20-169		
3	513	5FA	250A-WA-FR1-N-4566-01	1 OF 1	0	CJ-M15W14-PR-A20-169		
4	513	5FA	350A-WA-FR1-N-4518-01	1 OF 1	0	CJ-M15W14-PR-A20-169		
5	513	5FA	350A-WA-FR1-N-4519-01	1 OF 1	1	CJ-M15W14-PR-A20-169		
6	513	RFA	250A-WA-FR1-N-4562-01	1 OF 1	0	CJ-M15W14-PR-A20-000102/169		
7	513	RFA	250A-WA-FR1-N-4563-01	1 OF 1	0	CJ-M15W14-PR-A20-000103/169		
8	513	RFA	250A-WA-FR1-N-4564-01	1 OF 1	0	CJ-M15W14-PR-A20-000104/169		
9	513	RFA	250A-WA-FR1-N-4565-01	1 OF 1	0	CJ-M15W14-PR-A20-000105/169		
10	513	RFA	350A-WA-FR1-N-4518-01	1 OF 2	0	CJ-M15W14-PR-A20-169		
11	513	RFA	350A-WA-FR1-N-4518-02	2 OF 2	0	CJ-M15W14-PR-A20-169		
12	513	2FA	250A-WA-FR1-N-4566-01	1 OF 1	0	CJ-M15W14-PR-A20-169		
13	513	2FA	250A-WA-FR1-N-4571-01	1 OF 1	0	CJ-M15W14-PR-A20-169		
14	513	2FA	250A-WA-FR1-N-4572-01	1 OF 1	0	CJ-M15W14-PR-A20-169		



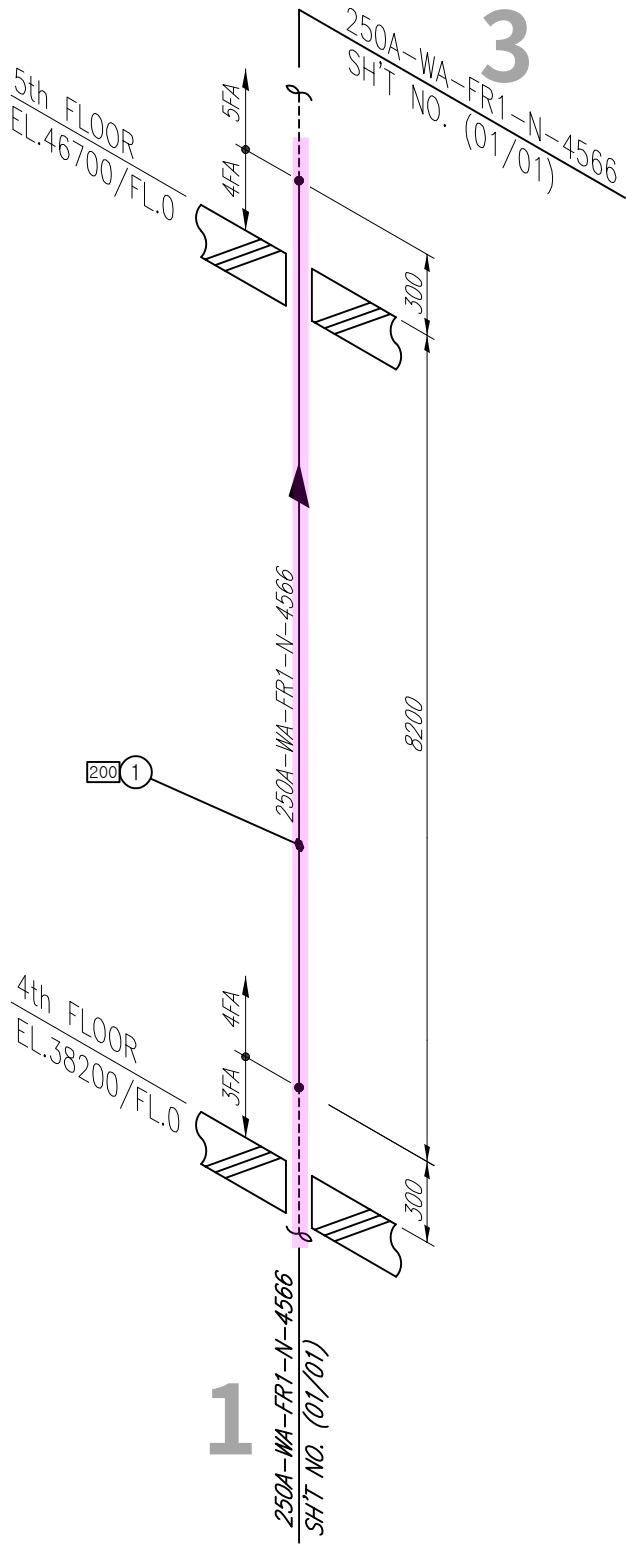
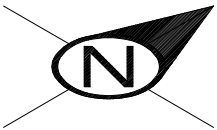
대외비

KEY PLAN (RF)

NOTE

1. 운전 시 CLOSE 되는 배관에는 LOTO 설치.

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정 REV.	실명 DESCRIPTION	날짜 DATE
제도 DRAWN BY이인학		
검토 CHECKED BY한진희		
승인 APPROVED BY박동영		
제작사 SK 에코플랜트		
공사명 PROJECT TITLE M15X Ph4 WWT Project		
도면명 DWG. TITLE PIPING & INSTRUMENT DIAGRAM ROOF FAN		
축척 SCALE NONE		
날짜 DATE 2025. 08. 14		
PROJECT NO. 22330D		REV.
도면번호 SHEET NO. CJ-M15W14-PR-A20-000169		1A



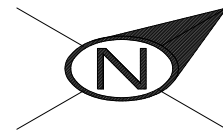
<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR		PROJECT NAME		
	THICK.	—	DESIGN	1.0	AMB	UNIT	4FA									M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY		JOB NO.		
						REF. P & ID CJ-M15W14-PR-A20-000169								WA		22330D		
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	250A-WA-FR1-N-4566		SH'T NO. 01 / 01	REV. 0



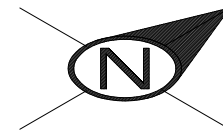
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1. (*) FIELD ADJUST

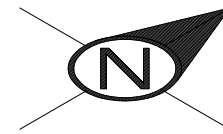
CLIENT  3

CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY WA		JOB NO. 22330D	
LINE NO. 250A-WA-FR1-N-4566	SH'T NO. 01 / 01		REV. 0

[illegible]

CLIENT				4			
							
CONTRACTOR				PROJECT NAME			
				M15X Ph4 Project			
COMMODITY				JOB NO.			
WA				22330D			
LINE NO.				SH'T NO.		REV.	
350A-WA-FR1-N-4518				01 / 01		0	

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1 * STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 * GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH * STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1												
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	-	DESIGN	1	AMB	UNIT 5FA						
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050009-C		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
						REF. P & ID CJ-M15W14-PR-A20-000169						
						REV. NO. DATE			DESCRIPTION	BY	CHK	APP



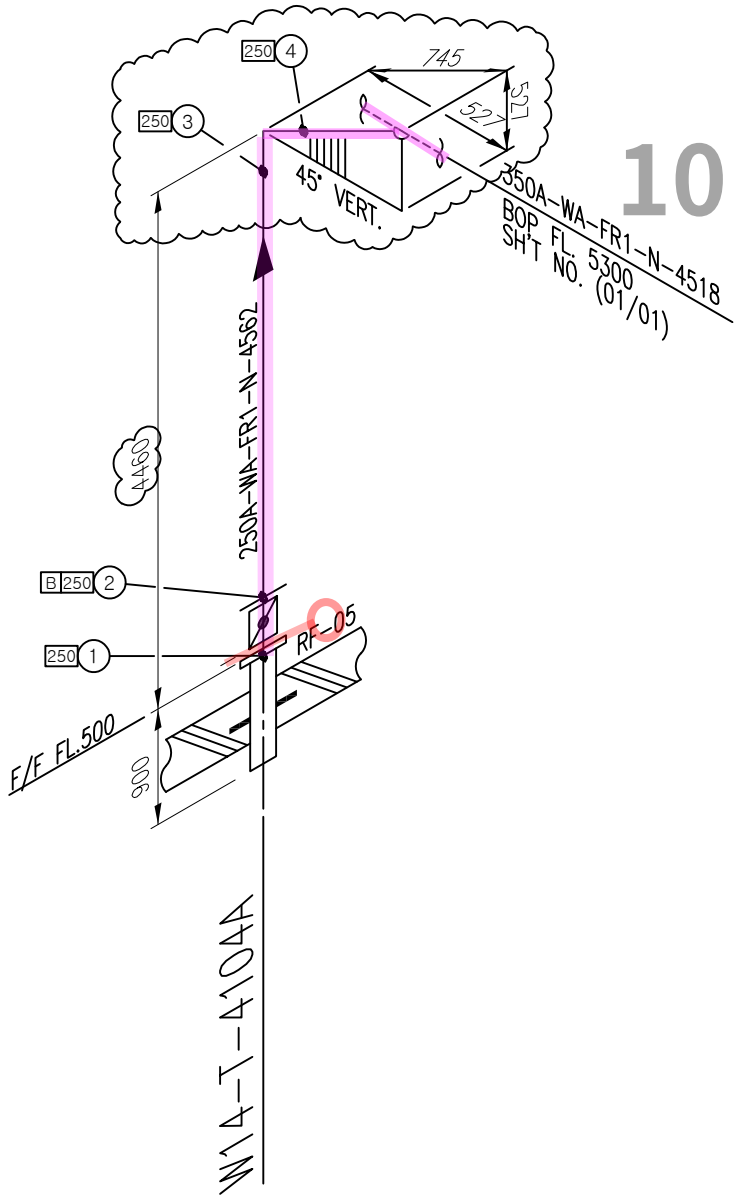
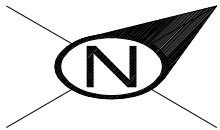
1. (*) FIELD ADJUST

[illegible]

CLIENT  5

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1.0	AMB	UNIT 5FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050005-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영		COMMODITY	WA
								25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			22330D
						REF. P & ID CJ-M15W14-PR-A20-000169	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	350A-WA-FR1-N-4519	SH'T NO. 01 / 01

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출을 할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



10

<NOTE>

1. (*) FIELD ADJUST



ITEM	SIZE	Q'TY	DESCRIPTION
FR1			
P	250A	5.2M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
E90	250A	3EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
F	250A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	250A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	250A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
VP	250A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER
E45	250A	1EA	ELBOW 45 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T

CLIENT



6

CONTRACTOR



PROJECT NAME

M15X Ph4 Project

COMMODITY

WA

JOB NO.

22330D

LINE NO.

250A-WA-FR1-N-4562

SH'T NO.

01 / 01

REV.

0

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION

CODE

N

CONDITION

PRESS.
(kg/cm2g)

TEMP.
(°C)

AREA

W14

THICK.

-

DESIGN

1

AMB

UNIT

RFA

OPERATING

ATM

AMB

REF. PLAN DWG.
CJ-M15W14-PI-A30-R10008-A

REF. P & ID
CJ-M15W14-PR-A20-000169



REV.
NO.

25.01.24

APPROVED FOR CONSTRUCTION

황래중

안남후

박동영

DATE

DESCRIPTION

BY

CHK

APP'D

CONTRACTOR



PROJECT NAME

M15X Ph4 Project

COMMODITY

WA

JOB NO.

22330D

LINE NO.

250A-WA-FR1-N-4562

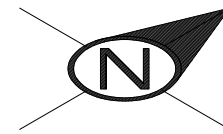
SH'T NO.

01 / 01

REV.

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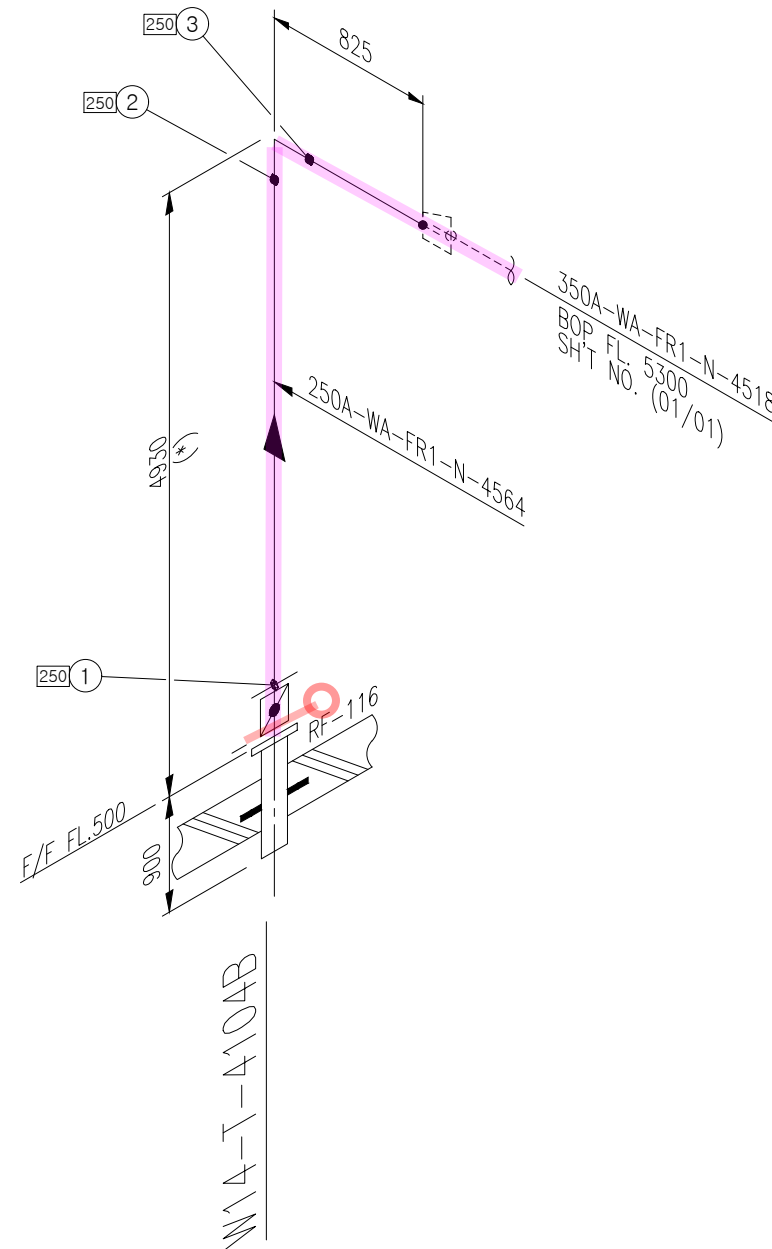
경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



1. (*) FIELD ADJUST

7

경고 : 본 문건은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단으로 복제, 제3자에게 공개, 누설 또는 사용함으로써 회사에 손해가 발생할 경우 응징 조치될 수 있으며, 회사는 응징 조치를 취할 권리가 있습니다.

[illegible]

10

<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VBLO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

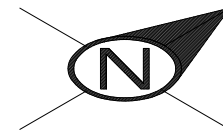
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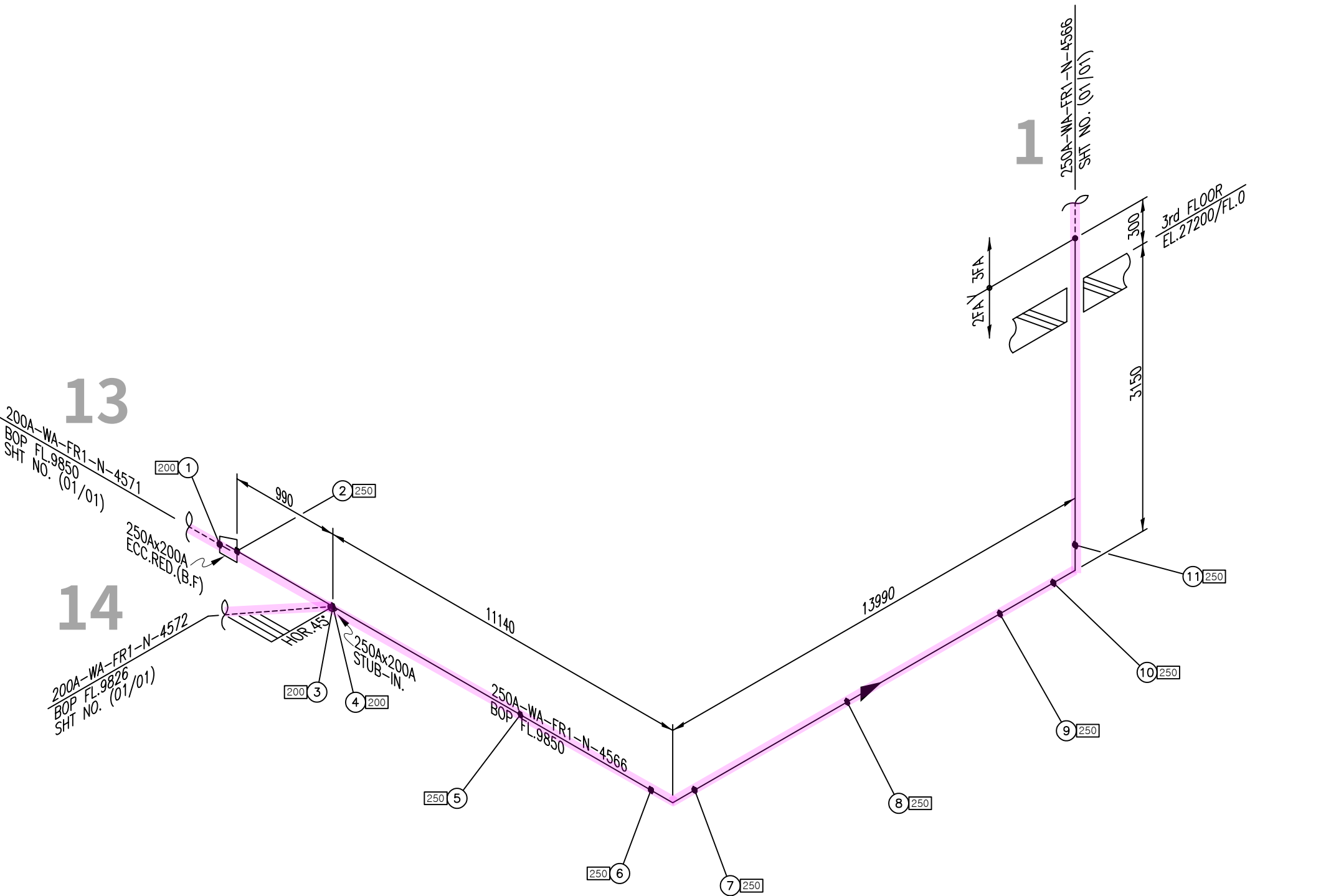
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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR 	PROJECT NAME M15X Ph4 Project	
	THICK.	—	DESIGN	1	AMB	UNIT RFA									
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10009-A	 REV. NO.	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY WA	JOB NO. 22330D	
						REF. P & ID CJ-M15W14-PR-A20-000169		DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO. 250A-WA-FR1-N-4564	SH'T NO. 01 / 01	REV. 0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



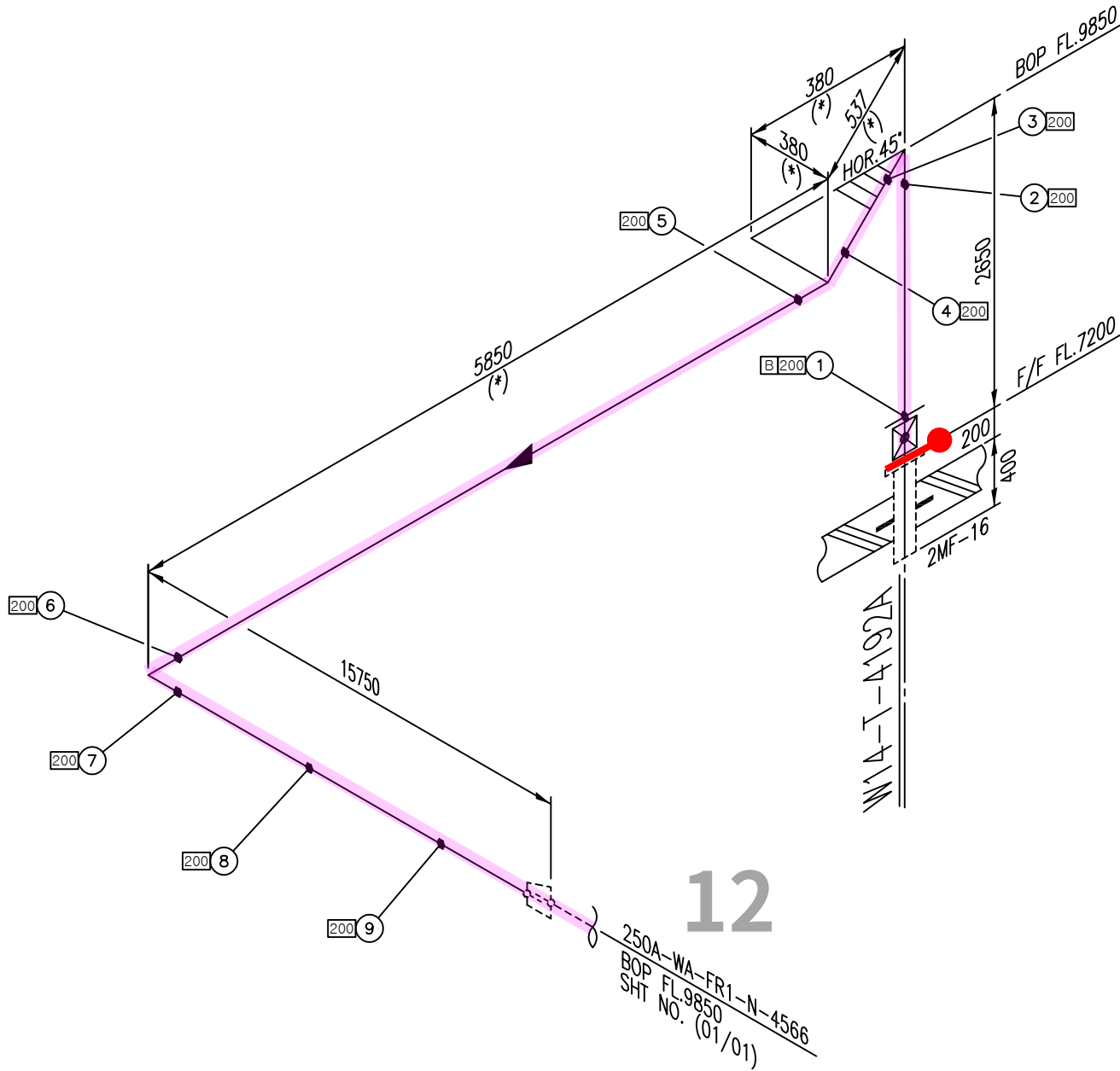
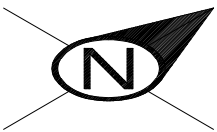
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY	WA	JOB NO.	
		22330D	
LINE NO.	350A-WA-FR1-N-4518	SH'T NO.	REV.
		02 / 02	0



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL EPDM LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	1	AMB	UNIT	2FA										M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-020011-E			24.12.20	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			COMMODITY	JOB NO.		
						REF. P & ID CJ-M15W14-PR-A20-000169											WA		22330D
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			LINE NO.	250A-WA-FR1-N-4566	SH'T NO. 01 / 01	REV. 0



<NOTE>

1. (*) FIELD ADJUST



ITEM	SIZE	Q'TY	DESCRIPTION
FR1			
P	200A	24.8M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 3.2T
E90	200A	2EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2T
E45	200A	1EA	ELBOW 45 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2T
VP	200A	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER
F	200A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
G	200A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	200A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

CLIENT



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CONTRACTOR	PROJECT NAME
SK ecoplant	M15X Ph4 Project
COMMODITY	JOB NO.
WA	22330D
LINE NO.	SH'T NO. REV.
200A-WA-FR1-N-4571	01 / 01 0

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL EPDM LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1	AMB	UNIT	2FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.								COMMODITY	WA	
						CJ-M15W14-PI-A30-020011-E			24.12.20	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영		JOB NO.	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	22330D	
						CJ-M15W14-PR-A20-000169								200A-WA-FR1-N-4571	SH'T NO.	REV.
															01 / 01	0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.

[illegible]

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* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA : VG/VG1/VG2	* BALL VA : VB/VB1/VB10/VB2/VB20	* BUTTERFLY VA : VF/VF4	* CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA : VL/VL1/VL2	* DIAPHRAGM VA : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL EPDM LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

[illegible]